$G \leq GL(n,\mathbb{C})$ is a MLG. A one parameter subgroup of $G$ is a smooth group homomorphism $\varphi : (\mathbb{R},+) \to G$

$\hookrightarrow \varphi(\varepsilon_1+\varepsilon_2) = \varphi(\varepsilon_1)\,\varphi(\varepsilon_2)$

$\varphi(0) = \mathbb{1}$

Turns out that $\varphi(\varepsilon)$ must be $e^{\varepsilon X}$ for some $X \in M_n(\mathbb{C})$

$$\varphi'(\varepsilon) = \frac{d}{d\varepsilon_1}\varphi(\varepsilon+\varepsilon_1)\Big|_{\varepsilon_1=0} = \frac{d}{d\varepsilon_1}\varphi(\varepsilon)\varphi(\varepsilon_1)\Big|_{\varepsilon_1=0} = \varphi(\varepsilon)\,\varphi'(0)$$

$$\begin{cases} \varphi'(\varepsilon) = \varphi(\varepsilon)\,\varphi'(0) \\ \varphi(0) = \mathbb{1} \end{cases} \rightarrow \varphi(\varepsilon) = e^{\varepsilon\varphi'(0)}$$

Why ops? Infinitesimal: connect a given group element to the identity

if $\varphi$ is a o.p.s. $\varphi'(0) \in \mathfrak{g}$ $\varphi'(\varepsilon)$ $\varepsilon \neq 0$ belongs to tangent space at $\varphi(\varepsilon)$

$\downarrow$

tangent space at $\mathbb{1} = \varphi(0)$

---

$$\underline{u}(n) = \left\{ X \in M_n(\mathbb{C}) \;\middle|\; e^{\varepsilon X} \in U(n) \quad \forall \varepsilon \in \mathbb{R} \right\}$$

$\rightarrow (e^{\varepsilon X})^* e^{\varepsilon X} = \mathbb{1} \quad \forall \varepsilon$

$e^{\varepsilon X^*} = e^{-\varepsilon X} \quad \forall \varepsilon$

$X^* = -X$ anti-Hermitian

$$\underline{u}(n) = \left\{ X \in M_n(\mathbb{C}) \;\middle|\; X^* = -X \right\}$$

$$\underline{su}(n) = \left\{ X \in M_n(\mathbb{C}) \;\middle|\; X^* = -X,\ tr(X) = 0 \right\}$$

$\underline{u}(1) = i \cdot \mathbb{R}$

Lie algebras are <u>real</u> vector spaces

- we can find the <u>dimension</u> of Lie algebra $\rightarrow$ by def $\dim(G) = \dim(\mathrm{Lie}(G))$ ← real dimension
- we can find a basis

<u>ex:</u> $X \in u(n)$ $\quad X \in M_n(\mathbb{C})$, $X^* = -X$ $\quad \frac{n^2-n}{2}$

$X = A + iB \quad A, B \in M_n(\mathbb{R})$

$A^T - iB^T = -A - iB \rightarrow$ $A^T = -A \rightarrow \frac{n(n-1)}{2}$ d.o.f

$B^T = B \rightarrow \frac{n(n+1)}{2}$ d.o.f

$= \frac{n(n-1)}{2} + \frac{2n}{2}$

total d.o.f: $n^2$

$$\boxed{\dim(U(n)) = n^2}$$

$\dim O(n) = \frac{n(n-1)}{2}$ $\quad \dim SO(n) = \dim O(n)$ because $\underline{so}(n) \equiv \underline{o}(n$

$\dim GL(n, \mathbb{R}) = n^2$

$\dim GL(n, \mathbb{C}) = 2n^2$

$\dim(SU(n)) = ?$ $\quad tr(X) = 0 \quad X = A + iB$

$0 = tr(A + iB) = \cancel{tr(A)}^{\,0} + i\, tr(B) = i\, tr(B) = 0$

remove 1 d.o.f.

<u>ex:</u> $\underline{SU(2)} = \{ X \in M_2(\mathbb{C}) \mid X^* = -X,\ tr(X) = 0 \}$

$X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ $\quad \begin{pmatrix} \bar{a} & \bar{c} \\ \bar{b} & \bar{d} \end{pmatrix} = \begin{pmatrix} -a & -b \\ -c & -d \end{pmatrix} \rightarrow$ $a = ix,\ x \in \mathbb{R}$

$d = iy,\ y \in \mathbb{R}$

$\bar{c} = -b$

set $b = u + iv$

$c = -\bar{b} = -u + iv$

$X = \begin{pmatrix} ix & u + iv \\ -u + iv & iy \end{pmatrix}$

$tr(X) = 0 \iff ix + iy = 0 \qquad y = -x$

$$\underline{su(2)} = \left\{ \begin{pmatrix} ix & u+iv \\ -u+iv & -ix \end{pmatrix} \;\middle|\; x, u, v \in \mathbb{R} \right\} \rightarrow \dim(su(2)) = 3$$

$$\boxed{\dim su(n) = n^2 - 1}$$

basis for $\underline{su(2)}$ : $\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \quad \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$

$\rightarrow$ generate $SU(2)$

another example: $SO(2) = \left\{ \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \;\middle|\; \theta \in \mathbb{R} \right\}$

$$\underline{so(2)} = \left\{ X \in M_2(\mathbb{R}) \;\middle|\; X^T = -X \right\}$$

$X = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \qquad \begin{pmatrix} a & c \\ b & d \end{pmatrix} = \begin{pmatrix} -a & -b \\ -c & -d \end{pmatrix} \qquad \begin{matrix} a = d = 0 \\ c = -b \end{matrix}$

$$\underline{so(2)} = \left\{ \begin{pmatrix} 0 & \varepsilon \\ -\varepsilon & 0 \end{pmatrix} \;\middle|\; \varepsilon \in \mathbb{R} \right\}$$

basis $\left\{ \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \right\}$ $\rightarrow$ generates all rotations

$$e^{\theta \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$$

$$\left\{ \begin{pmatrix} a & b \\ -b & a \end{pmatrix} \;\middle|\; a, b \in \mathbb{R} \right\} \cong \mathbb{C} \qquad a + ib \leftrightarrow \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$$

$$e^X = \mathbb{1} + X + \frac{X^2}{2} + \frac{X^3}{3!} + \dots$$

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \qquad J^2 = -\mathbb{1} \qquad J^{2n} = (-1)^n \mathbb{1}$$

$$J^3 = J^2 J = -J \qquad J^{2n+1} = (-1)^n J$$

$$e^{\theta J} = \sum_{k=0}^{\infty} \frac{\theta^k}{k!} J^k = \sum_{l=0}^{\infty} \frac{\theta^{2l} J^{2l}}{(2l)!} + \sum_{l=0}^{\infty} \frac{\theta^{2l+1} J^{2l+1}}{(2l+1)!}$$

$$= \underbrace{\sum_{l=0}^{\infty} \frac{\theta^{2l} (-1)^l}{(2l)!}}_{\cos\theta} \mathbb{1} + \underbrace{\sum_{l=0}^{\infty} \frac{\theta^{2l+1} (-1)^l}{(2l+1)!}}_{\sin\theta} J$$

$$= \cos\theta\, \mathbb{1} + \sin\theta\, J = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$$

$$\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} = \mathbb{1} + \underline{\underline{\begin{pmatrix} 0 & \theta \\ -\theta & 0 \end{pmatrix}}} + O(\theta^2)$$

---

<u>$SO(3)$</u> generators:

$$J_z = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad J_x = \begin{pmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad J_y = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$e^{\theta J_z} = \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$ rotation around z-axis

$$e^{\theta J_x} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{pmatrix}$$ rotation around x-axis

---

<u>Representations</u> goal: (linear) action of $G$ on a complex vector space $V$

$$G \times V \to V \qquad \mathbb{1} \triangleright v = v$$

$$(g, v) \mapsto g \triangleright v \qquad g_1 \triangleright g_2 \triangleright v = (g_1 g_2) \triangleright v$$

(left action)

<u>linear</u> $\quad g \triangleright (\alpha v_1 + \beta v_2) = \alpha (g \triangleright v_1) + \beta (g \triangleright v_2)$

$P = M \times G$

$V$ fields: $M \to M \times V$

$G$ acts on both $P$ and $V$

$U(1)$ gauge

$\alpha : M \to \mathbb{R}$ gauge field

$e^{\alpha(x)} \in G$ $\quad M \to G$

representation

$\psi : M \to \mathbb{C}$

action of $U(1)$ on $\mathbb{C}$ : $e^{i\theta} \triangleright z = e^{i\theta} z$

$\psi(x) \to e^{\alpha(x)} \psi(x)$ gauge transformation

$\mathcal{L} = \frac{1}{2} \nabla \bar{\psi} \nabla \psi$

$\nabla \psi \to e^{\alpha} \nabla \alpha \, \psi(x) + e^{\alpha} \nabla \psi$

$D \psi = \nabla \psi - e \alpha \psi$

another rep. $\quad e^{i\theta} \triangleright z = e^{i 2\theta} z$